A Synopsis
On
CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM
Phase I — Project Proposal

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING
(Machine Learning & Artificial Intelligence)

Submitted by:

Divyansh Uniyal (Lead) 2318766 Section ML2
Bhavya Gogia 2318662 Section C1
Km Shivani 2318999 Section ML2
Abhishek Negi 2318178 Section C1

Under the Guidance of

Ms. Neha Pokhriyal
Assistant Professor

Project Team ID: CSE27-293

Department of Computer Science and Engineering
Graphic Era Hill University
Dehradun, Uttarakhand

August-2026

CERTIFICATE

This is to certify that the project synopsis entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted by Divyansh Uniyal (2318766), Bhavya Gogia (2318662), Km Shivani (2318999), Abhishek Negi (2318178) for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the project synopsis has not been submitted in any other university or institution for the award of any degree.

Ms. Neha Pokhriyal
Assistant Professor
Department of Computer Science and Engineering,
Graphic Era Hill University, Dehradun, India.

CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project synopsis entitled "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM" is being submitted for the partial fulfillment of the requirement for the award of the degree of B.Tech C.S.E. (Machine Learning & Artificial Intelligence) in Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

Name of Student	Roll Number	Signature
Divyansh Uniyal (Lead)	2318766	
Bhavya Gogia	2318662	
Km Shivani	2318999	
Abhishek Negi	2318178	

TABLE OF CONTENTS

Sr. No.	Description	Page No.
	Abstract	5
1	Introduction	6
2	Literature Review	9
3	Problem Statement & Objectives	12
4	Proposed Methodology	15
5	Hardware and Software Requirements	26
6	Expected Outcomes	28
7	Conclusion	31
	References	32

Note: page numbers are field codes. Press Ctrl+A then F9 in Word to refresh them after any edit.

ABSTRACT

Urban traffic in Indian cities is heterogeneous and not lane disciplined: two-wheelers, auto-rickshaws, e-rickshaws, cars, buses and occasionally cattle share the same carriageway and occupy lateral positions that a lane model does not describe. Signals at such junctions are still overwhelmingly fixed-time, so green is allocated to an approach that may be empty while a saturated approach waits. The information needed to do better is already present in the CCTV cameras that most junctions already carry; what is missing is a method that turns those frames into a forecast a controller can act on.

This project proposes MFSTNet: a camera-only pipeline that predicts per-lane congestion sixty seconds ahead and hands that prediction to a reinforcement-learning signal controller. A fine-tuned YOLOv8s detector produces per-lane vehicle counts over eight India-relevant classes; those counts are converted into ordered congestion labels using passenger-car-unit-weighted occupancy with per-camera thresholds. A frozen convolutional backbone and a frozen self-supervised vision transformer are fused by gated bidirectional cross-attention, modelled over time by a bidirectional LSTM with temporal self-attention, and read out per lane. A PPO agent trained in the SUMO microsimulator consumes the forecast as part of its observation and selects the next phase and its duration.

The methodology is deliberately staged. Phase I delivers the corpus, the detector and a model with standard cross-attention only, and nothing from Phase II begins until that converges. Phase II adds gating, temporal attention and attention pooling, each behind a configuration flag so that an eight-configuration ablation — including a linear probe as its floor — runs without a code edit. Evaluation is leave-one-camera-out at five seeds, and the controller is measured against a swept, best-configuration Webster baseline over thirty paired seeds.

The expected outcome is a curated dataset, a fine-tuned detector, a trained forecasting model with its fusion gate exposed as an inspectable artefact, a signal controller benchmarked with confidence intervals and an effect size, and a working edge-to-server prototype in which no raw frame ever leaves the capture device. Every table in the final report is to be generated from committed per-run result files by a committed script.

Keywords: traffic congestion forecasting; cross-attention fusion; vision transformer; reinforcement learning; adaptive signal control; unstructured traffic. Working name: cnn-vit-bilstm-cross-attention-traffic-system.

CHAPTER 1

INTRODUCTION

1.1 Background

A signalised junction allocates a scarce resource — green time — among competing approaches. A fixed-time signal allocates it from a plan computed once, from counts taken once, and then applies that plan identically at every hour of every day. When demand is stable this is adequate. When demand varies, it is not: the plan is right on average and wrong most of the time, and the cost is paid in waiting vehicles, in fuel, and in emissions.

The Indian case is harder than the textbook case in two specific ways. First, the traffic stream is heterogeneous: a two-wheeler, an auto-rickshaw and a bus occupy very different amounts of road and discharge at very different rates, so counting vehicles without weighting them measures the wrong quantity. Second, the stream is not lane disciplined; vehicles occupy lateral positions between nominal lanes, and queue geometry does not follow lane markings. Both facts are recognised in Indian practice — the Indian Highway Capacity Manual [12] supplies passenger-car-unit equivalents precisely because vehicle counts are not comparable across classes — and both are ignored by most published traffic-vision work.

1.2 Motivation

Two observations motivate this project. The first is that the sensing problem is largely already solved by existing infrastructure: a large number of Indian junctions already carry fixed CCTV cameras, installed for enforcement and monitoring, whose frames are sufficient to count and classify vehicles per approach. Adding inductive loops or probe-vehicle feeds would improve the input, but it would also make the method deployable only where a municipality has already invested in them.

The second is that a controller does not need to know the present; it needs to know the near future. A phase decision taken now takes effect over the next thirty to ninety seconds, so a controller acting on the current queue is always acting on stale information. A forecast one signal cycle ahead is therefore not a refinement of the sensing problem but a different problem, and it is the one this project takes as central.

1.3 Selection of the Project Idea

The team shortlisted five candidate problems before fixing on one. They were scored against four criteria: whether every required input could be obtained without a dependency outside the team's control; whether the work contained a component that could be ablated and therefore defended; whether it fitted a twenty-week schedule with a zero-rupee cash budget; and whether a negative result would still be a reportable finding rather than a failed project. Table 1.1 records the shortlist and the decision on each.

Table 1.1 Candidate project ideas considered before selection

Candidate idea	What it would have required	Decision and reason
Helmet and traffic-violation detection from CCTV	Detection only; a solved problem with several public baselines.	Rejected — no forecasting or control component, so the project would end at a detector and have nothing to ablate.
Vehicle re-identification across junctions	Needs synchronised multi-camera footage with overlapping fields of view.	Rejected — the footage the team can obtain is single-view and unsynchronised, so the core assumption fails before any modelling.
Accident and anomaly detection	Positives are rare, and severity labels need expert adjudication.	Rejected — the positive class would be too small to evaluate honestly within a twenty-week schedule.
Sensor-fusion congestion forecasting (camera + loop detector + GPS)	Strongest formulation, but depends on municipal sensor feeds.	Rejected — no route to the loop-detector or probe-vehicle data; the project would stall on a dependency outside the team's control.
Camera-only congestion forecasting driving an adaptive signal controller	One fixed camera per approach, a short forecast horizon, and a controller trained in simulation against that forecast.	Selected — every input is obtainable from a single fixed camera, and the work splits cleanly into a perception claim and a control claim that can each stand alone.

The selected idea was registered under the working name cnn-vit-bilstm-cross-attention-traffic-system, which names the pipeline rather than the contribution: a convolutional branch, a vision-transformer branch, cross-attention between them and a recurrent temporal model. That name is retained here because it is how the idea appears in the team's early records, and guideline §1 asks for identity fields to be traceable across documents. The title carried forward to every later document is "CNN VIT BILSTM CROSS-ATTENTION BASED TRAFFIC MANAGEMENT SYSTEM".

It is worth being explicit about what is claimed. The individual mechanisms in this pipeline are published: cross-attention, gated fusion, frozen self-supervised backbones and PPO signal control all exist in the literature. The contribution claimed is their combination for

this specific setting, and three things that are usually left out of it — the fusion gate reported as an artefact rather than an internal detail, a strictly camera-only input, and an evaluation that holds out whole cameras. Overstating novelty in a proposal only creates a claim that has to be retracted later.

1.4 Scope of Phase I

This synopsis proposes the whole project, but the work is gated. Phase I is the mandatory core: the corpus pipeline, the detector, and a forecasting model built from a convolutional branch, a transformer branch, standard cross-attention, a bidirectional LSTM and a per-lane head. The architectural refinements that make the work publishable are Phase II, and they are not begun until the Phase I model trains to convergence. This ordering is a deliberate guard against a known failure mode in student projects — an elaborate architecture with no experiments attached to it — and it is treated here as a constraint rather than a preference.

1.5 Organisation of this Synopsis

Chapter 2 reviews the relevant literature in four strands and states the gap. Chapter 3 states the problem, the research questions, the numbered objectives and the scope boundaries. Chapter 4 gives the proposed methodology: the design decisions and what they were chosen over, the system architecture, the corpus, the model, the controller, the evaluation protocol, the phase plan and the risk register. Chapter 5 lists the hardware and software the project actually requires. Chapter 6 states the expected outcomes, the measurable targets and the future scope. Chapter 7 concludes.

CHAPTER 2

LITERATURE REVIEW

The work this project builds on falls into four strands: visual representation for traffic scenes, multimodal fusion, temporal modelling for short-horizon forecasting, and adaptive signal control. Each is summarised below in terms of what it establishes and what it leaves unresolved for the present setting.

2.1 Visual Representation for Traffic Scenes

Deep residual networks [4] remain the default feature extractor for dense scenes, and their inductive bias suits the problem: local texture, partial occlusion and small objects are exactly what a convolutional stack is good at. The limitation is equally structural. A convolutional receptive field is local, and the quantity of interest — how far a queue extends along an approach — is global. A network that sees the head of a queue clearly and the tail not at all will underestimate it consistently.

Vision transformers [2] model global relationships by construction, and DINOv2 [3] shows that self-supervised pre-training produces features that transfer without fine-tuning. That last property is what makes a transformer branch affordable here: if the backbone is never updated, its features can be computed once and cached, and the training cost collapses. What is not established is how well frozen self-supervised features describe dense, unstructured traffic specifically, since almost all reported transfer results fine-tune. Treating that as an open question rather than an assumption is why the count-only baseline is part of this experiment.

2.2 Multimodal and Multi-Branch Fusion

Cross-attention [1] is the standard mechanism for letting one representation query another, and gated combinations of two branches are common in multimodal work. The mechanisms are therefore not novel, and this proposal does not claim them as such. What is unusual is the reporting: fusion is normally evaluated by whether the fused model scores higher, and the internal quantity that decides how much of each branch was used is not published. A gate that is logged, tracked and analysed turns that quantity into evidence, and makes it possible to say which branch the model relied on under which conditions rather than merely that fusion helped.

2.3 Temporal Modelling for Short-Horizon Forecasting

Long short-term memory networks [14] are long established for sequence modelling, and gradient-boosted trees [15] remain strong on tabular count sequences. Sixty timesteps is a short sequence: long enough that a recurrence has something to accumulate, short enough that a transformer trained from scratch on a small corpus would be a poor use of the available data. The combination adopted here — a bidirectional recurrence followed by a small self-attention block over its outputs — is chosen on those grounds and is itself part of the ablation rather than an assumption.

2.4 Adaptive Signal Control

Webster's method [10] computes cycle length and green splits from flow ratios and remains the classical baseline. Its two parameters matter more than its formula: the saturation flow rate, for which the published range for Indian conditions spans more than a factor of two [11], [12], and the cycle-length clamp. A learned controller compared against one arbitrary point in that range is being compared against a straw baseline.

Reinforcement learning for signal control is a mature subfield; the survey of Wei et al. [13] catalogues both its progress and its recurring methodological problems, of which weak baselines and single-seed reporting are the two most common. PPO [7], for which a maintained implementation is available [8], is stable on small discrete action spaces without the replay tuning that value-based methods require; the underlying formulation follows standard treatment [16]. Training requires a microsimulator, and SUMO [9] is the established open one.

2.5 Comparative Summary

Table 2.1 places the strands beside one another, stating for each what it establishes and what it leaves open for this project.

Table 2.1 Comparative summary of the four literature strands

Strand	What it establishes	What it leaves open here
Convolutional backbones for traffic scenes	Deep residual networks [4] give strong local texture and occlusion cues, and remain the default feature extractor for dense scenes.	Local receptive fields alone do not represent the whole approach; a queue that extends past the crop is not visible to them.
Vision transformers and self-supervised features	ViT [2] models global context by construction, and DINOv2 [3] shows self-supervised features transfer without fine-tuning.	Almost all reported use is fine-tuned. How well frozen DINOv2 features describe dense unstructured traffic is not established.

Strand	What it establishes	What it leaves open here
Cross-attention fusion	Cross-attention [1] is the standard way to let two modalities query one another, and is used widely in vision-language work.	Fusion is normally reported as a single number. Which branch the model actually relied on is not reported, so it cannot be checked.
Temporal modelling for short-horizon forecasting	Recurrent models [14] and gradient-boosted count models [15] are both established for short-horizon traffic prediction.	Count-only baselines are frequently omitted from vision papers, so the appearance branch is rarely shown to be earning its place.
Adaptive signal control	Webster's method [10] remains the classical baseline; the RL survey [13] catalogues both the progress and the recurring evaluation flaws of learned controllers.	Learned controllers are often compared against an untuned fixed-time arm, which makes the improvement easy to overstate.
Unstructured-traffic datasets and standards	IDD [6] covers Indian road scenes for driving tasks; IRC [11] and Indo-HCM [12] supply the intersection and capacity conventions.	No public dataset gives fixed-camera junction footage with the per-lane congestion labels this project needs.

2.6 Research Gap

Three gaps follow from the above and together define this project. There is no published fixed-camera junction dataset for unstructured traffic carrying per-lane congestion labels; the closest Indian resource [6] is built for driving tasks from a moving platform, not for a fixed junction view. There is no reported analysis of what a fusion gate between a convolutional and a transformer branch actually learns on traffic imagery. And learned signal controllers are seldom evaluated against a baseline that has itself been tuned, which makes the reported improvement difficult to interpret.

CHAPTER 3

PROBLEM STATEMENT & OBJECTIVES

3.1 Problem Statement

Given only the video streams of fixed cameras at a signalised junction carrying heterogeneous, non-lane-disciplined traffic, predict the congestion level of each approach sixty seconds ahead, and use that prediction to select the next signal phase and its duration so that mean vehicle waiting time is lower than under a tuned fixed-time plan — subject to the constraint that no raw video frame is transmitted off the capture device or written to disk.

Three parts of that statement are load-bearing. "Only the video streams" excludes loop detectors and probe-vehicle data, which makes the method deployable on existing infrastructure and also makes it harder. "Sixty seconds ahead" is one signal cycle, which is the horizon over which a phase decision has effect. And the privacy constraint is a design constraint rather than a policy note: it determines that inference runs at the edge and that only derived counts and predictions cross the network.

3.2 Research Questions

The project is organised around three questions, each stated so that a negative answer is a reportable result rather than a failure.

RQ1 Do frozen appearance features add anything over vehicle counts alone when forecasting per-lane congestion sixty seconds ahead? The count-only baseline is part of the experiment, not a courtesy.

RQ2 Does gated bidirectional fusion beat cheaper fusion, and does the learned gate carry an interpretable signal about which branch the model relies on?

RQ3 Does a controller that receives a short-horizon congestion forecast outperform the same controller without one, measured against a swept-and-tuned Webster baseline rather than an untuned one?

3.3 Objectives

The objectives below are ordered by dependency: each requires the one before it. Objectives O1 to O3 constitute Phase I and are mandatory; O4 and O5 are Phase II; O6 is Phase III.

O1 Assemble a licence-audited corpus of fixed-camera junction footage covering unstructured traffic, and fine-tune a YOLOv8s detector over eight vehicle classes that include the auto-rickshaw, e-rickshaw and cattle absent from COCO.

O2 Derive per-lane congestion labels from detector counts using PCU-weighted occupancy with per-camera thresholds, so that a label means the same thing at a six-lane arterial and a two-lane approach.

O3 Implement the Phase I model — two frozen backbones, standard cross-attention fusion, a BiLSTM and a per-lane congestion head — and demonstrate that it trains to convergence before any Phase II component is added.

O4 Extend the model with gated bidirectional fusion, temporal self-attention and attention pooling, and quantify each addition through an eight-configuration ablation that includes a linear probe as its floor.

O5 Evaluate leave-one-camera-out, reporting mean and bootstrap confidence interval across folds together with the seed spread, so that no reported score is a memorised camera or a lucky initialisation.

O6 Train a PPO signal controller on a SUMO intersection whose observation vector carries the forecast, measure it against a swept, best-configuration Webster baseline over thirty paired seeds, and demonstrate an edge-to-server prototype in which no raw frame leaves the capture device.

3.4 Scope

The following are within scope and are what the project undertakes to deliver.

- One four-approach signalised junction, with one fixed camera per approach and a fixed field of view.
- A sixty-second forecast horizon at a five-second sampling interval, giving a sixty-step input window.
- Three ordered congestion classes per lane — LOW, MEDIUM, HIGH — defined on PCU-weighted occupancy rather than on raw vehicle count.
- Signal control evaluated in simulation, with the prototype demonstrating the end-to-end path rather than controlling live signals.

3.5 Out of Scope

The following are explicitly excluded. They are listed rather than omitted, because an unstated exclusion reads as an unfulfilled promise at assessment.

- Control of a live municipal signal. Nothing in this project is connected to a road-side controller.
- Vehicle re-identification, number-plate recognition, or any per-vehicle identity. Only aggregate counts leave the capture device.
- Network-wide coordination across multiple junctions. The observation and action spaces are defined for one junction.
- Pedestrian and non-motorised phase modelling beyond a fixed minimum clearance interval.
- Night-time and heavy-rain conditions, which are recorded as a stated limitation rather than claimed and left untested.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Design Decisions and the Alternatives Rejected

Every component below was chosen from a shortlist, and the shortlist is recorded rather than discarded. Table 4.1 gives, for each decision, what was considered, what was selected and the reason. Two of the entries are forced by the operating envelope rather than by preference — the frozen backbones by 6 GB of available video memory, and the edge device by a zero-rupee cash budget — and are marked as such rather than presented as principled choices they are not.

Table 4.1 Design decisions taken at project inception, with the alternatives rejected

Component	Alternatives considered	Selected	Reason
Detector	Faster R-CNN; RetinaNet; DETR; YOLOv8s	YOLOv8s	Single-stage and exportable, and the only candidate with a credible path to the ten-frames-per-second edge budget.
Vision transformer	Supervised ViT-S/16; CLIP ViT-B/32; DINOv2 ViT-S/14	DINOv2 ViT-S/14	The backbones are frozen, so representation quality is the whole of what the branch contributes. Note the patch size: 14, not 16, so token counts must be read from configuration rather than assumed.
Backbone regime	Full fine-tuning; partial unfreeze partway through training; frozen backbones with a precomputed feature cache	Frozen with a cache	At the specified batch size an uncached step pushes 1,920 frames through both backbones, which does not fit in 6 GB. Caching turns the eight-configuration ablation from days into hours.
Fusion mechanism	Channel concatenation; one-way cross-attention; bidirectional cross-attention averaged; gated bidirectional cross-attention	Gated bidirectional	The gate is a scalar per position that says which branch the model relied on. It is the one part of the design that produces an inspectable artefact rather than only a score.
Temporal model	GRU; temporal convolutional network; transformer encoder alone; BiLSTM followed by temporal self-attention	BiLSTM + temporal self-attention	Sixty steps is short for a transformer trained from scratch and long enough that the recurrence has something to carry.

Component	Alternatives considered	Selected	Reason
Sequence pooling	Last hidden state; mean over time; attention pooling	Attention pooling	Congestion sixty seconds ahead is not equally determined by every frame, and attention pooling is the ablation-friendly way to say so.
Prediction target	Per-frame count regression; one congestion class for the junction; per-lane three-class congestion	Per-lane three-class at t + 60 s	The controller acts per approach, so a junction-level label would be unusable by the component that consumes it.
Lane geometry	Automatic clustering of detections; homography from a survey; hand-placed centres reviewed by a second person	Hand-placed and reviewed	Automatic centres move by 0.164 of frame width when the detector changes, which silently changes every count the corpus is built on.
Evaluation split	Random 60/20/20; stratified by class; leave-one-camera-out	Leave-one-camera-out	A random split lets the model memorise a camera and report it as generalisation. Holding out a whole camera is the only split that answers the question actually being asked.
Controller	Fixed-time; vehicle-actuated; DQN; PPO	PPO	Stable on a small discrete action space without the replay-buffer tuning DQN needs, and available as a maintained implementation.
Simulator	CityFlow; SUMO with the default lane model; SUMO with the sublane model	SUMO, sublane model	Indian traffic is not lane-disciplined. The sublane model represents lateral position within a lane; the residual gap is declared rather than papered over.
Edge device	NVIDIA Jetson Nano; Raspberry Pi with a Coral accelerator; a team laptop	Team laptop	The cash budget is zero. The control logic and the message contract are unchanged by the substitution, but every latency figure must then state the host it was measured on.
Datastore	PostgreSQL with TimescaleDB; InfluxDB; SQLite	SQLite	The prototype stores a bounded history of counts and decisions. A database server would be infrastructure without a purpose.

4.2 Proposed System Architecture

The system is three tiers, shown in Figure 4.1. The edge tier reads frames, detects and counts vehicles per lane, and publishes counts only. The server tier runs the forecasting model and the controller, applies safety limits, and stores a bounded history. The presentation tier displays live values and benchmark values in separate panels that never merge, so that a number measured on a held-out split is never shown as though it were a live reading.

The tier boundary carries the project's privacy constraint. Frames are read, processed and discarded inside the edge tier; what crosses the network is a count vector, a prediction and a signal command. Two fallback paths are required behaviour rather than robustness extras: if the forecasting model is unavailable the controller runs on counts alone, and if the message broker is silent for more than ten seconds the edge node arms a locally embedded fixed-time controller.

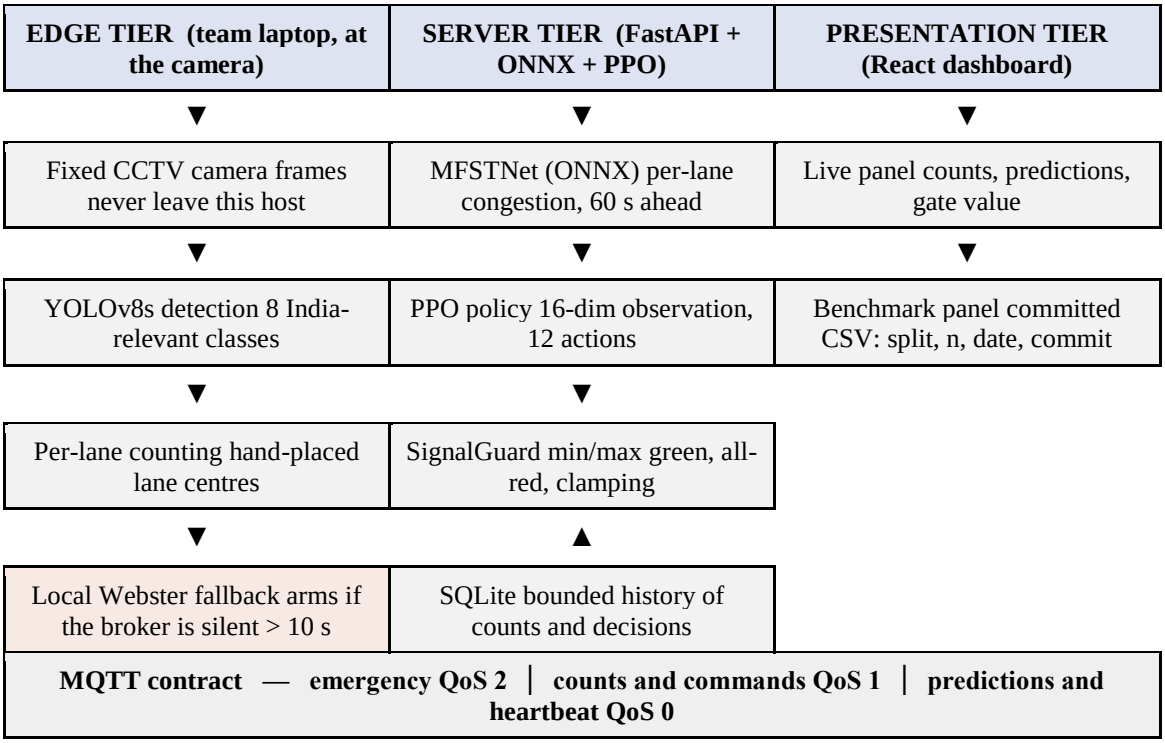


Fig. 4.1 Proposed three-tier system architecture

4.3 Dataset and Corpus Construction

No public dataset supplies fixed-camera junction footage with per-lane congestion labels, so the corpus is constructed. The dataset is two-track by design. A detector is bootstrapped from public unstructured-traffic imagery [6] so that the pipeline is never blocked waiting on in-house annotation, and an in-house set is curated and captured in parallel and swapped in when ready. Any experiment run in the interval records which detector weights produced it, because two sets exist during that window.

Figure 4.2 gives the pipeline. Two stages deserve comment. Lane geometry is placed by hand and reviewed rather than derived automatically, because automatically derived lane centres shift by roughly one sixth of the frame width when the detector changes, which would silently alter every count the corpus rests on. And the split is cut by source clip, never by window: overlapping temporal windows from one clip placed on both sides of a split would leak the answer.

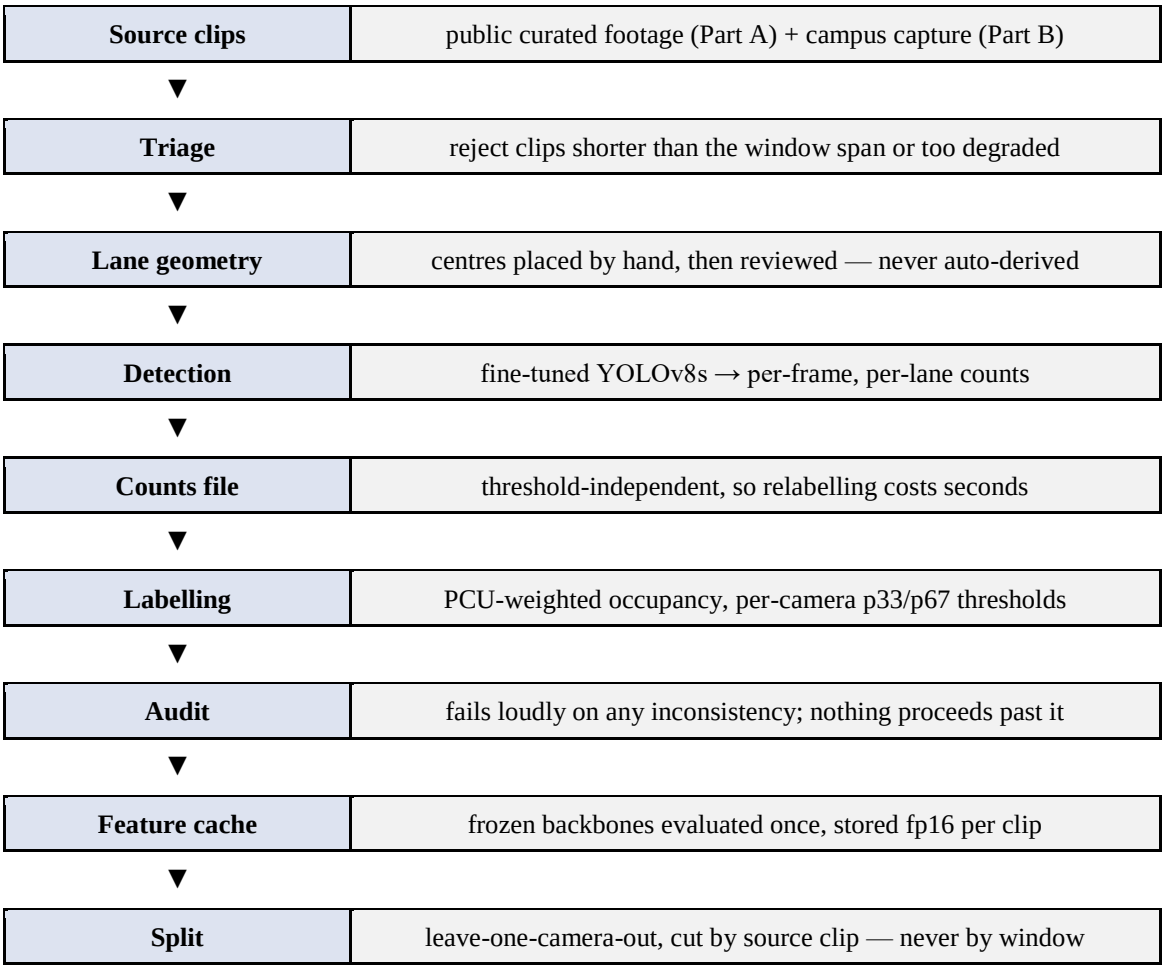


Fig. 4.2 Proposed corpus construction pipeline

4.3.1 Labelling by PCU-Weighted Occupancy

A congestion label derived from a raw vehicle count would treat a lane holding ten two-wheelers and a lane holding ten buses as equally congested. Labels are therefore defined on passenger-car-unit-weighted occupancy, using the equivalences published in the Indian Highway Capacity Manual [12]. For lane l at time t, with n_{c,l}(t) detections of class c and pcu_c the equivalence factor for that class, equation (4.1) defines the occupancy that thresholds are applied to.

$$occ_l(t) = (SUM_c \; n_{c,l}(t) \cdot pcu_c) \; / \; cap_l \quad (4.1)$$

Here cap_l normalises by the capacity of the lane, so that the quantity is comparable between a six-lane arterial approach and a two-lane one. Thresholds separating LOW from MEDIUM and MEDIUM from HIGH are calibrated per camera at the 33rd and 67th percentiles of that camera's own occupancy distribution and are recorded in the corpus manifest. A vehicle class with no published equivalence raises an error rather than defaulting to a factor of one, since a silent default would corrupt every label containing that class.

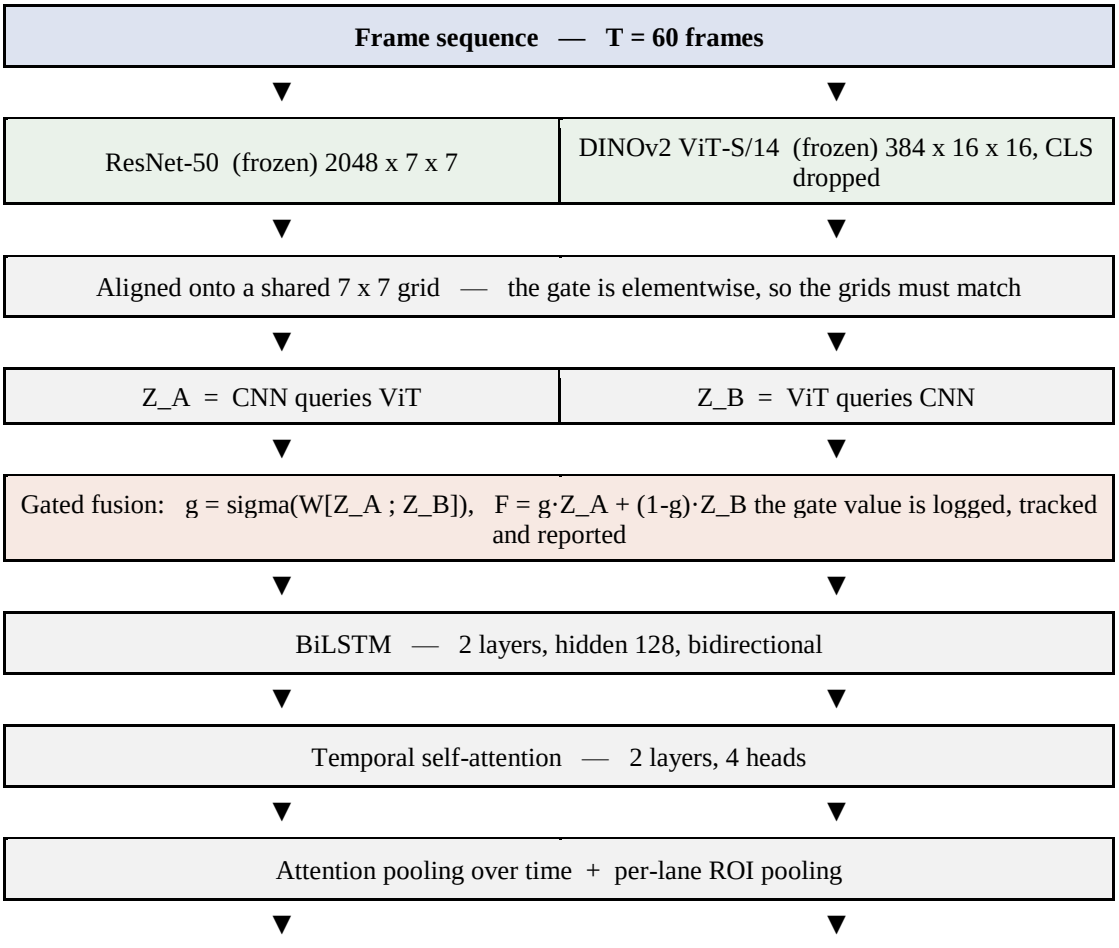
4.4 Vehicle Detection and Counting

Detection uses YOLOv8s [5], fine-tuned over eight classes chosen for the Indian street: two-wheeler, auto-rickshaw, e-rickshaw, car, light commercial vehicle, bus, truck and cattle. The last three of those are either absent from or unreliable in general-purpose object-detection training sets, which is the reason fine-tuning is necessary rather than optional. The detection split is stratified 70/15/15, which is a different split from the model evaluation described in Section 4.7 and serves a different purpose.

Detection quality and counting quality are reported separately. A detector can score well on mean average precision while systematically miscounting a class that is small in the frame, and since every congestion label in the corpus is derived from counts, it is the counting error that propagates. The counting operating point is therefore calibrated against human-verified counts on a held-out set, and never against the detector's own output.

4.5 The Proposed Forecasting Model

The model takes a sixty-frame window sampled at five-second intervals and predicts, for each of the four approaches, one of three ordered congestion classes at sixty seconds after the end of the window. Its structure is given in Figure 4.3.



Per-lane head — LOW / MEDIUM / HIGH for N, S, E, W at t + 60 s

Fig. 4.3 Proposed model structure

4.5.1 Frozen Backbones

Both backbones — a residual convolutional network [4] and a self-supervised vision transformer [3] — are frozen. This is not a regularisation choice; it is what makes the project fit the hardware. With both backbones trainable, a single step at the specified batch size would push 1,920 frames through them, which does not fit in 6 GB. Frozen, their outputs are a deterministic function of the input, so they are computed once and cached, and the eight-configuration ablation becomes a matter of hours rather than days. The cache is keyed on the preprocessing configuration and refuses to load against a mismatched key, because silently reusing features computed under different normalisation would produce results that are wrong rather than merely stale.

4.5.2 Gated Bidirectional Cross-Attention

Two cross-attention passes [1] are computed: Z_A, in which the convolutional features query the transformer features, and Z_B, in which the transformer features query the convolutional features. A learned gate then combines them, as in equations (4.2) and (4.3), where sigma is the logistic function and W and b are learned.

$$g = \mathrm{sigma}(W [Z_A ; Z_B] + b) \quad (4.2)$$

$$F_{fused} = g \cdot Z_A + (1 - g) \cdot Z_B \quad (4.3)$$

The gate value g is treated as an output of the system rather than an internal detail. It is returned from the forward pass, logged during training, displayed on the dashboard and analysed in the final report, because it is the quantity that answers which branch the model relied on. A gate that is computed and discarded would make the fusion claim untestable.

4.5.3 Temporal Modelling and Read-Out

The fused per-timestep features pass through a two-layer bidirectional LSTM [14], then through a small temporal self-attention block, and are pooled over time by attention rather than by taking the last hidden state — congestion sixty seconds ahead is not equally determined by every frame in the window. Read-out is per lane, using region-of-interest pooling over the spatial grid at each lane's reviewed geometry. Pooling globally and then applying one shared head four times cannot produce four different predictions, which is why the feature cache preserves spatial structure rather than storing a pooled vector.

4.5.4 Planned Ablation

Every component above is required to be disableable by configuration, giving the eight configurations in Table 4.2. Configuration H, a linear probe over mean-pooled frozen features, is the floor: any frozen-backbone architecture that cannot beat it has not justified itself, and it is reported whatever it shows.

Table 4.2 The eight planned ablation configurations

Config	Variant	What it isolates
A	CNN only	ResNet-50 features alone; no ViT, no fusion.
B	ViT only	DINOv2 ViT-S/14 features alone; no CNN, no fusion.
C	Concatenation	Both backbones, channel concatenation instead of attention.
D	One-way cross-attention	CNN queries ViT only; the reverse path is disabled.
E	Bidirectional, no gate	Both directions, averaged rather than gated.
F	Gated + temporal attention	Adds temporal self-attention over the BiLSTM outputs.
G	Full model	Gated bidirectional fusion, temporal attention, attention pooling.
H	Linear probe	Frozen features, mean over time, linear classifier. The floor.

4.6 Adaptive Signal Control

The controller is a PPO agent [7], [16] implemented with a maintained library [8], acting on a four-way intersection modelled in the SUMO microsimulator [9] and driven through its online interface. Geometry and phase structure follow Indian intersection design practice [11]. Because the traffic being modelled is not lane disciplined, the simulator's sublane model is enabled so that vehicles may occupy lateral positions within a lane; the residual difference between the simulation and observed behaviour is to be declared as a limitation rather than left implicit.

The observation vector is sixteen-dimensional and its layout is a contract: per-lane queue lengths, current phase and elapsed green, time of day, an emergency flag, and four entries carrying the forecast for the four approaches. If the forecast is unavailable those four entries are zeroed and the vector keeps its length, because a shortened vector silently invalidates every trained checkpoint. The action space is twelve discrete actions: two phase groups by six green durations. The reward, equation (4.4), is the negative mean waiting time across lanes at each decision step.

$$r_t = -\left(\frac{1}{|L|}\right) \sum_{l \in L} w_l(t) \quad (4.4)$$

The baseline is Webster's method [10], and it is swept rather than configured once: the published saturation-flow range for Indian conditions is wide enough that a single arbitrary value would produce whichever result the author preferred. The best Webster configuration found by that sweep is the baseline the learned controller must beat.

4.7 Evaluation Plan and Performance Measures

The forecasting model is to be evaluated leave-one-camera-out: each fold holds out one camera entirely, with the remainder used for training and validation. A random split over windows would let the model recognise a camera it has already seen and report that as generalisation. The headline figure is the mean across folds with a bootstrap 95% confidence interval, quoted together with the number of folds, because an interval over a handful of folds is not an interval over thirty runs.

Every configuration is to be run at five seeds. A single-seed table ranks initialisations rather than architectures, and on a corpus of this size the movement between seeds can exceed the gap between two architectures. The evaluation harness therefore reports the seed spread beside every mean, and two configurations separated by less than that spread are to be reported as not separated by the experiment.

Count-only baselines are part of the experiment rather than a courtesy. A recurrent model and a gradient-boosted model [15] over the same count sequences, on the same folds and seeds, answer RQ1 directly: if the appearance branch does not beat them, that is the finding and it will be reported as such. Table 4.3 lists the measures to be reported and why each is included.

Table 4.3 Performance measures and the reason for each

Measure	Why it is reported
Macro-F1	The headline forecasting metric. Chosen over accuracy because the HIGH class is expected to be rare, and accuracy would reward ignoring it.
Per-class precision, recall and support	Reported for every class, so a metric cannot hide an unpredicted class.
Quadratic weighted kappa and ordinal error rate	The classes are ordered: confusing LOW with HIGH is not the same error as confusing LOW with MEDIUM.
Detection mAP@0.5 and counting error	Reported separately, because counting accuracy and detection accuracy are different quantities and the corpus depends on the second.
Mean vehicle waiting time and queue length	The control metrics, measured over thirty paired evaluation seeds per method in each demand regime.

Measure	Why it is reported
Bootstrap 95% confidence interval, paired t-test, Cohen's d	Every comparison is reported with an interval and an effect size, never as a bare difference of two means.
Latency, with its measurement host	Detection frame rate, model inference time, controller decision time and message round-trip, each labelled with the machine it was measured on.

Three reproducibility rules apply throughout. Seeds are fixed across the numerical, deep-learning and reinforcement-learning libraries and recorded with every result. Raw per-run results are committed, not only summary tables. And every table in the final report is generated from those committed files by a committed script, so that a number in the report and a number in the data cannot disagree.

4.8 Phase-Wise Development Plan

The build order is a constraint. Table 4.4 states the three phases, what each contains and the condition that must hold before the next begins. The rule that matters is the gate between Phase I and Phase II: no gating, no temporal attention and no attention pooling is implemented until the Phase I model trains cleanly to convergence. Figure 4.4 shows the same sequence as a flow, with the gate conditions on the transitions.

Table 4.4 Phase-wise development plan and exit conditions

Phase	Weeks	Contents	Exit condition
Phase I Mandatory	Weeks 1-8	Corpus pipeline and audit; YOLOv8s fine-tuning; PCU-weighted labelling; feature cache; CNN + ViT with standard cross-attention, BiLSTM and the per-lane head.	The Phase I model trains to convergence and beats the linear probe on a held-out camera. Nothing from Phase II begins until it does.
Phase II Stretch	Weeks 9-14	Gating replaces standard cross-attention; temporal self-attention over the BiLSTM outputs; attention pooling replaces the last hidden state; the full eight-configuration ablation, leave-one-camera-out at five seeds.	Every configuration runs from a flag with no code edit between configurations, and the ablation table is generated from committed per-run CSVs.
Phase III Optional	Weeks 15-20	SUMO environment and PPO arms; the thirty-seed paired benchmark against a swept Webster baseline; ONNX export; edge node, MQTT contract, control server and dashboard.	The prototype demonstrates both required fallback paths, and the control result is reported with its disqualification rules applied.

PHASE I mandatory	Corpus + detector + labels -> CNN + ViT + standard cross-attention + BiLSTM -> per-lane head
▼	GATE: the Phase I model trains to convergence and beats the linear probe on a held-out camera
PHASE II stretch	Gating replaces standard cross-attention -> temporal self-attention -> attention pooling -> ablation A-H, leave-one-camera-out at five seeds
▼	GATE: every configuration runs from a flag, and the ablation table is generated from committed per-run results
PHASE III optional	SUMO + PPO arms -> thirty-seed benchmark against a swept Webster baseline -> ONNX export, edge node, control server, dashboard

Fig. 4.4 Build order, with the condition on each gate

Table 4.5 places those phases on the twenty-week calendar as dated milestones, each of which is a checkable state of the repository rather than an activity.

Table 4.5 Twenty-week milestone schedule

Milestone	State to be reached
Week 2	Idea fixed, requirements written, detector bootstrapped on a public dataset so the corpus pipeline is not blocked on in-house annotation.
Week 5	Corpus pipeline end to end: triage, reviewed lane geometry, counts, labels, audit. Architecture and design documents due.
Week 8	Phase I model converging; feature cache built; the in-house dataset swaps in and every experiment records which detector weights produced it.
Week 11	Phase II components in; ablation running; test plan and test design documents due.
Week 14	Leave-one-camera-out ablation complete at five seeds, with raw per-run CSVs committed.
Week 17	Controller trained and benchmarked; ONNX export and control server working.
Week 19	Prototype integrated end to end, both fallback paths exercised.
Week 20	Results, test report and paper draft; public code release with pinned requirements and fixed seeds.

4.9 Risk Register

Table 4.6 is the risk register. Each entry names a risk, its likelihood and impact, and the response — not a hope. Two entries are worth reading together: the possibility that the appearance features do not beat a count-only baseline, and the possibility that the forecast does not help the controller. Both are treated as outcomes to be measured rather than as failure modes to be avoided, which is why the two claims are kept separate and why the count-only baseline is in the experiment from the first week.

Table 4.6 Risk register with likelihood, impact and planned response

Risk	Likelihood	Impact	Response
Annotation effort is underestimated	High	High	Bootstrap the detector on a public dataset from Week 2 so the corpus pipeline never waits on in-house annotation, and curate from permissively licensed sources before collecting anything new.
Permission for campus footage is delayed	Medium	High	The perception track is built on curated public footage first, so in-house capture swaps in as an addition rather than a dependency.
A capture location yields only one camera	Medium	Medium	Leave-one-camera-out cannot score a single-camera location. Capture is planned as multiple viewpoints per location from the start.
The HIGH class is rare in the available footage	High	Medium	Report macro-F1 and per-class support rather than accuracy, weight the loss by inverse frequency, and state the class counts beside every result.
6 GB of VRAM is insufficient	Medium	High	Freeze both backbones and cache their features; hold a small number of windows resident and accumulate gradients to the specified batch size.
The simulator does not reproduce unstructured traffic	High	Medium	Use the sublane model, calibrate demand against observed counts, and declare the residual gap as a limitation rather than claiming realism.
The forecast does not help the controller	Medium	High	Run the perfect-forecast ceiling first: if a perfect forecast does not beat no forecast, the forecast is not worth carrying and that is reported as a finding.
Appearance features do not beat counts	Medium	High	The count-only baseline is in the experiment from the start, and the two claims are stated separately so the control claim ships even if the vision claim does not.

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

The requirements below are the ones this project actually needs; nothing is listed because it is conventional. Table 5.1 gives the hardware and Table 5.2 the software, each entry stated with the purpose it serves rather than as a specification in isolation.

5.1 Hardware Requirements

Two entries determine the shape of the whole project. The 6 GB of video memory on the training GPU is why both backbones are frozen and their features cached, and the substitution of a team laptop for dedicated edge hardware is why every latency figure in this project must state the host it was measured on.

Table 5.1 Hardware requirements

Component	Specification	Purpose in this project
Training workstation	Laptop, 8-core x86-64 CPU, 16 GB RAM	Feature caching, model training and the ablation sweeps.
GPU	NVIDIA RTX 4050 Laptop, 6 GB VRAM	Frozen-backbone feature extraction and training, in mixed precision. The 6 GB limit is why the backbones are frozen and cached.
Storage	512 GB SSD, of which roughly 60 GB for video, cache and weights	Source clips, the memory-mapped feature cache and model checkpoints.
Edge node	A team laptop acting as the edge device	Detection, counting and the local fallback controller. Every latency figure states this host, because it is not a Jetson.
Camera	Fixed municipal or campus CCTV stream, 720p or better	Frame source, one per approach. Frames are processed in place and never transmitted.
Network	Wi-Fi or Ethernet LAN with a broker inside 200 ms	Carries counts, predictions and signal commands only.

5.2 Software Requirements

Exact versions are to be pinned in a committed requirements file, which is the authority; the groups in Table 5.2 state what is required and why. One installation detail is recorded here because it has a failure mode that is silent: the CUDA-enabled build of the deep-learning framework must be installed from its own package index before the remaining

packages, since the default index resolves to a CPU-only build that then fails at run time rather than at install time.

Table 5.2 Software requirements

Category	Software
Operating system and language	Windows 11 and Ubuntu 22.04 (both supported); Python 3.12+
Deep learning	PyTorch with CUDA 12.6; torchvision, for the ResNet-50 backbone; timm, for the DINOv2 backbone; Ultralytics YOLOv8
Reinforcement learning and simulation	Stable-Baselines3; Gymnasium; Eclipse SUMO with TraCI and sumolib
Serving and deployment	ONNX and ONNX Runtime; FastAPI with Pydantic; Uvicorn; Eclipse Mosquitto with Paho-MQTT
Data, storage and analysis	SQLite, embedded, with no separate database server; NumPy, pandas and SciPy; scikit-learn and XGBoost; OpenCV
Interface	React with Vite and Zustand; Recharts and native WebSocket
Engineering tooling	Git, with Git LFS for model weights; pytest with continuous integration; MLflow and TensorBoard for experiment tracking; Docker Compose

CHAPTER 6

EXPECTED OUTCOMES

6.1 Deliverables

The project is expected to produce seven artefacts, listed in Table 6.1. Each is stated so that its completion is checkable: an artefact that exists on disk, in a repository or in a release, rather than a description of work done.

Table 6.1 Expected deliverables

Deliverable	Description
Curated dataset	Fixed-camera junction footage curated from permissively licensed sources, plus an in-house capture set, with a per-camera provenance record stating how each property was determined.
Fine-tuned detector	YOLOv8s weights over eight India-relevant classes, with the counting operating point calibrated against human-verified counts rather than against its own pseudo-labels.
Congestion forecasting model	Trained weights, an ONNX export for CPU inference, and the gate value exposed from the forward pass as a research artefact rather than an internal detail.
Ablation and evaluation results	Per-run CSVs for all eight configurations across every fold and seed, with confidence intervals generated by a committed script and never transcribed.
Signal controller	A trained PPO policy and a thirty-seed paired benchmark against a swept Webster baseline and a fixed-time arm, across three demand regimes.
Working prototype	Edge node, MQTT contract, control server with safety limits enforced outside the transport layer, and a monitoring dashboard that keeps benchmark numbers and live numbers in separate panels.
Reproducibility package	Pinned requirements, fixed seeds, committed raw results, a public code release and archived weights, so every reported number can be regenerated from a clean machine.

6.2 Measurable Targets

Table 6.2 states the criteria the finished work is to be judged against. They are targets set in advance, not predictions of the result: a target that is missed is to be reported as missed, with the measured value and an analysis, rather than quietly restated afterwards to match whatever was obtained.

Table 6.2 Measurable targets for the completed project

Aspect	Target	How it is to be measured
Detection quality	mAP@0.5 ≥ 0.70 over the eight classes	Measured on a human-labelled test split, never on pseudo-labels.
Detection throughput	≥ 10 frames per second on the edge device	Reported with the measurement host named, because the edge device is a laptop and not a Jetson.
Forecasting quality	Macro-F1 ≥ 0.75 on held-out cameras, and above the count-only baseline	Both halves matter: beating the baseline is the claim, the absolute number is the target.
Model inference	≤ 150 ms per window on server CPU via ONNX	The budget the control loop is designed around.
Control improvement	≥ 15% lower mean waiting time than the best Webster configuration	Measured over thirty paired seeds with a paired t-test and an effect size, and disqualified if fewer than 85% of trips complete.
Decision and message latency	≤ 50 ms per control decision and ≤ 200 ms end to end over MQTT	Measured on the prototype, with the host stated.
Reproducibility	Every table regenerated from committed CSVs by a committed script	Fixed seeds across the numerical, deep-learning and RL libraries.

6.3 Future Scope

The following extensions are outside the twenty-week schedule and are recorded so that the boundary of this project is explicit rather than implied.

Coordination across junctions — The observation and action spaces are defined for one junction. Extending to a corridor means a shared reward and an explicit communication scheme between agents, which is a project in itself.

Deployment on dedicated edge hardware — The control logic and the message contract are written to be unchanged by the substitution, so a Jetson port is a re-measurement rather than a rewrite.

Wider domain coverage — Night, rain and fog are out of scope here and are the first extension worth making, because they are where a camera-only method is weakest.

Distilling a larger detector — A larger detector can pseudo-label footage for the deployed small one, adopted only if pre-registered criteria are met on real labelled test data.

Longer forecast horizons — Sixty seconds is chosen to match one signal cycle. Longer horizons would let the controller plan across cycles, at a cost in accuracy that has to be measured rather than assumed.

Public release and dissemination — Code, weights and an archived dataset record, with a paper submitted to a transport or vision venue once the evaluation is complete.

CHAPTER 7

CONCLUSION

This synopsis proposes MFSTNet: a camera-only system that forecasts per-lane congestion one signal cycle ahead at a junction carrying heterogeneous, non-lane-disciplined traffic, and a reinforcement-learning controller that acts on that forecast. The perception component fuses a frozen convolutional branch and a frozen self-supervised transformer branch through gated bidirectional cross-attention, models sixty timesteps with a bidirectional recurrence and temporal attention, and reads out per lane. The control component is a PPO agent trained in microsimulation against a swept classical baseline.

Three commitments distinguish the proposal from a description of an architecture. The first is order: Phase I is a mandatory core that must converge before any refinement is written, which is a deliberate guard against an elaborate model with no experiments attached. The second is falsifiability: a count-only baseline and a linear probe are in the experiment from the beginning, the evaluation holds out whole cameras, every configuration is run at five seeds, and the classical control baseline is tuned before it is beaten. The third is that the fusion gate is reported as an artefact rather than kept as an internal detail, which is what makes the fusion claim checkable at all.

The risks are known and stated in Section 4.9 with responses attached, and the two most consequential — that appearance features may not beat vehicle counts, and that the forecast may not help the controller — are treated as measurements to be taken rather than outcomes to be avoided. The two claims are therefore kept separate, so that a null result on the perception question does not invalidate the control result or the prototype.

On completion the project is expected to deliver a curated dataset, a fine-tuned detector, a trained and exported forecasting model, a benchmarked controller, a working edge-to-server prototype in which no raw frame leaves the capture device, and a reproducibility package from which every reported number can be regenerated. Whether each component earns its place is a question the experiments are designed to answer honestly, in either direction.

REFERENCES

[1] A. Vaswani et al., "Attention is all you need," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2017, pp. 5998-6008.

[2] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. Int. Conf. Learn. Represent. (ICLR), 2021.

[3] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," Trans. Mach. Learn. Res., 2024.

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770-778.

[5] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[6] G. Varma, A. Subramanian, A. Namboodiri, M. Chandraker, and C. V. Jawahar, "IDD: A dataset for exploring problems of autonomous navigation in unconstrained environments," in Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV), 2019, pp. 1743-1751.

[7] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[8] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," J. Mach. Learn. Res., vol. 22, no. 268, pp. 1-8, 2021.

[9] P. A. Lopez et al., "Microscopic traffic simulation using SUMO," in Proc. IEEE Int. Conf. Intell. Transp. Syst. (ITSC), 2018, pp. 2575-2582.

[10] F. V. Webster, "Traffic signal settings," Road Research Technical Paper No. 39, Road Research Laboratory, London, U.K., 1958.

[11] Indian Roads Congress, IRC:SP:41-1994, Guidelines for the Design of At-Grade Intersections in Rural and Urban Areas. New Delhi, India: IRC, 1994.

[12] Indo-HCM, Indian Highway Capacity Manual. New Delhi, India: CSIR-Central Road Research Institute, 2017.

[13] H. Wei, G. Zheng, V. Gayah, and Z. Li, "Recent advances in reinforcement learning for traffic signal control: A survey of models and evaluation," ACM SIGKDD Explorations Newsletter, vol. 22, no. 2, pp. 12-18, 2021.

[14] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.

[15] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, 2016, pp. 785-794.

[16] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.